

be achieved using two or four copper pairs, and one such extension of SHDSL provides data rates up to 5.696 Mbps.

The payload may be either unstructured, T1, E1, multiple ISDN basic rate access (BRA), asynchronous transfer mode (ATM) cells or Ethernet packet transfer mode (PTM). In order to share the SHDSL bandwidth, a dual bearer mode can be used, which allows a combination of two types of payloads.

Asymmetric digital subscriber line (ADSL). By studying different scenarios, it was realised that it was possible to transmit data more quickly from an exchange to a user. But when the user sent information to the exchange, it was more sensitive to the noise caused by electromagnetic disturbances (the nearer the subscriber to the exchange, the greater the concentration of cables, generating more crosstalk). So the idea was to use an asymmetric system, imposing a lower speed from the subscriber to the exchange. This idea gave birth to the asymmetric digital subscriber line technology, which was originally developed at Bellcore (now Telcordia Technologies) in 1988.

ADSL caters specifically to connections between ISPs and customers. The Internet is used largely for downloading files, HTML and graphical content. Processes like uploading files or other content to servers are limited to very few users. Hence the bandwidth required for downstream data (from ISP to client) is more than that required for upstream data (from client to ISP).

This DSL-based technology enables transmission and reception of data at speeds higher than legacy copper media. The modulation technique used allows several bits to be represented by one transmission symbol.

In ADSL, bit rate allocation for a channel within the available bandwidth is not the same as for the other channels, and hence the term 'asymmetric.' In other words, the upstream bandwidth is smaller than the downstream bandwidth. ADSL offers an upstream data rate of 500 kbps and a downstream data rate of up to 8 Mbps.

ADSL Lite, another variant of the ADSL standard, offers upstream speeds up to 500 kbps and downstream speeds up to 1.5 Mbps. Further, ADSL has many variants like ADSL2, splitterless ADSL2, ADSL2+ and ADSL++.

ADSL2/G.DMT.bis is defined in ITU G.992.3 and is an improved version of ADSL with data rates of 12 Mbps in downstream and 3.5 Mbps in upstream. Splitterless ADSL2/G.lite.bis is defined in ITU G.992.4 and is capable of providing 1.536Mbps downstream and 512kbps upstream.

ADSL2+ defined in ITU G.992.5 can provide up to 24Mbps theoretical downstream speed, which is double of the ADSL2 speed. The upstream speed is up to 3.5 Mbps. Thus ADSL2+ doubles the frequency band of typical ADSL from 1.1 MHz to 2.2 MHz. More importantly, ADSL2+ provides port bonding known as G.998.x or G.Bond. This is a very attractive feature of ADSL2+ in which the download and upload speeds are the sum of individual speed of all provisioned ports to the end user. It means if two lines with 24Mbps were bonded, the net result would be a speed of 48 Mbps.

ADSL++, another variant of ADSL, developed by Centillium Communications, is capable of providing download speeds up to 50Mbps, and uses the frequency band up to 3.75 MHz.

Rate-adaptive digital subscriber line (R-ADSL). R-ADSL operates with the same transmission rates as ADSL, but the modem adjusts dynamically to varying lengths and quantities of the twisted-pair local lines. It makes possible to connect over different lines at varying speeds. Connection speed is negotiated by the end-points when the line synchronises up or as a result of a signal from the central office.

R-ADSL is designed to increase range and noise tolerance by sacrificing upstream speed. The modem automatically creates a greater frequency band for the downstream than the upstream band. If line noise or signal degradation is large, the upstream bandwidth is decreased and may fall up to 64 kbps,

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All the above Modules are Ready to Interface with any Micro controller

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Project Guidance also available